



Last-Mile Delivery Intelligence Platform

Supply Chain Analytics Consulting Engagement

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Executive Summary

PROBLEM

35.85% of deliveries failing 12-day SLA, representing major refund exposure.

FINDINGS

Premium products have worst OTIF (58.75%).
Northern states drive 90%+ failure rates.
Operations generated 659.47K kg in CO₂ emissions.

RECOMMENDATION

Implement ML-powered risk feed, redistribute courier capacity, and optimize shipment modes to reduce the 6.84 kg CO₂ per delivery average.

Current Delivery Performance

64.15%

OTIF Rate

35.85%

Failure Rate

12.09 days

Avg Delivery

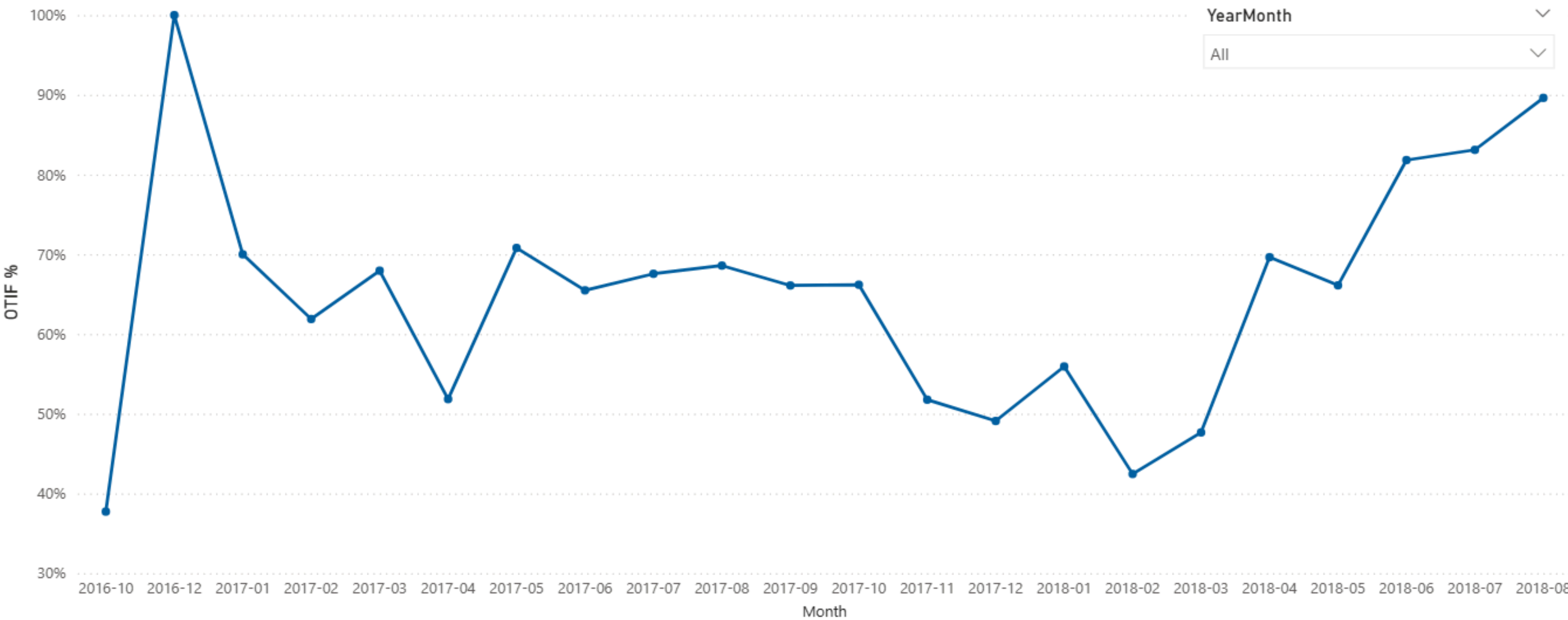
659.47K kg

Total CO₂

6.84 kg

Avg CO₂/Delivery

Monthly OTIF % Trend — Delivery Performance 2016-2018



Filter by Month

YearMonth

All

Root Cause: Where Are Deliveries Failing?

Zone Heatmap — Geographic Delivery Performance

State-level OTIF analysis across Brazil — identifying high-risk delivery zones

SP

State with the Best OTIF

AL

State with the Worst OTIF

22

States Below 50%

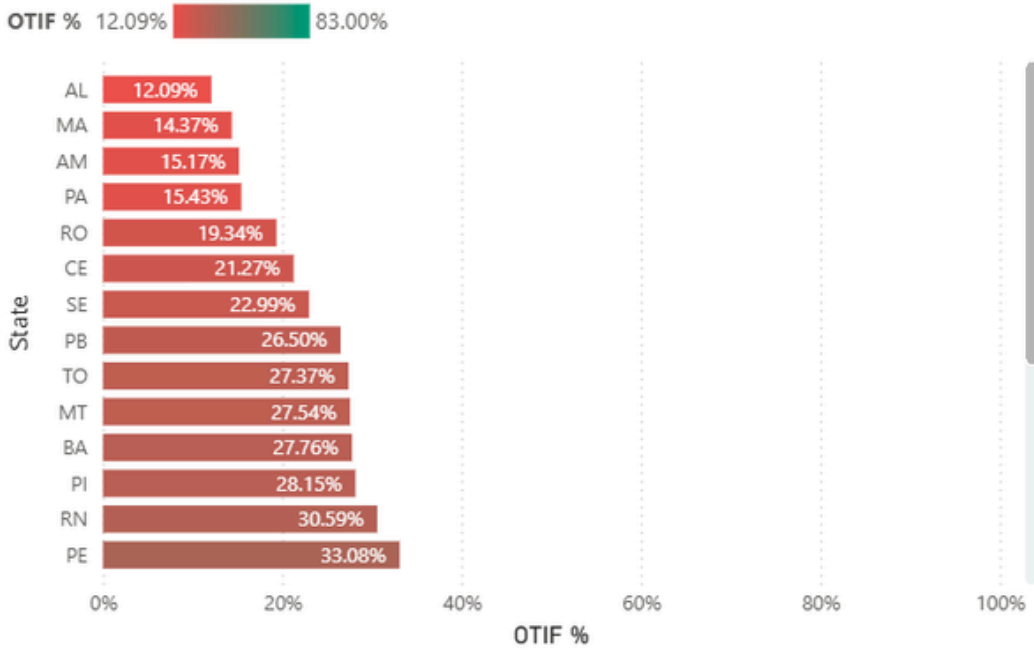
⚠ This visual type is being retired soon. Contact your admin to upgrade.

Upgrade map

OTIF % by Brazilian State



State Performance Ranking



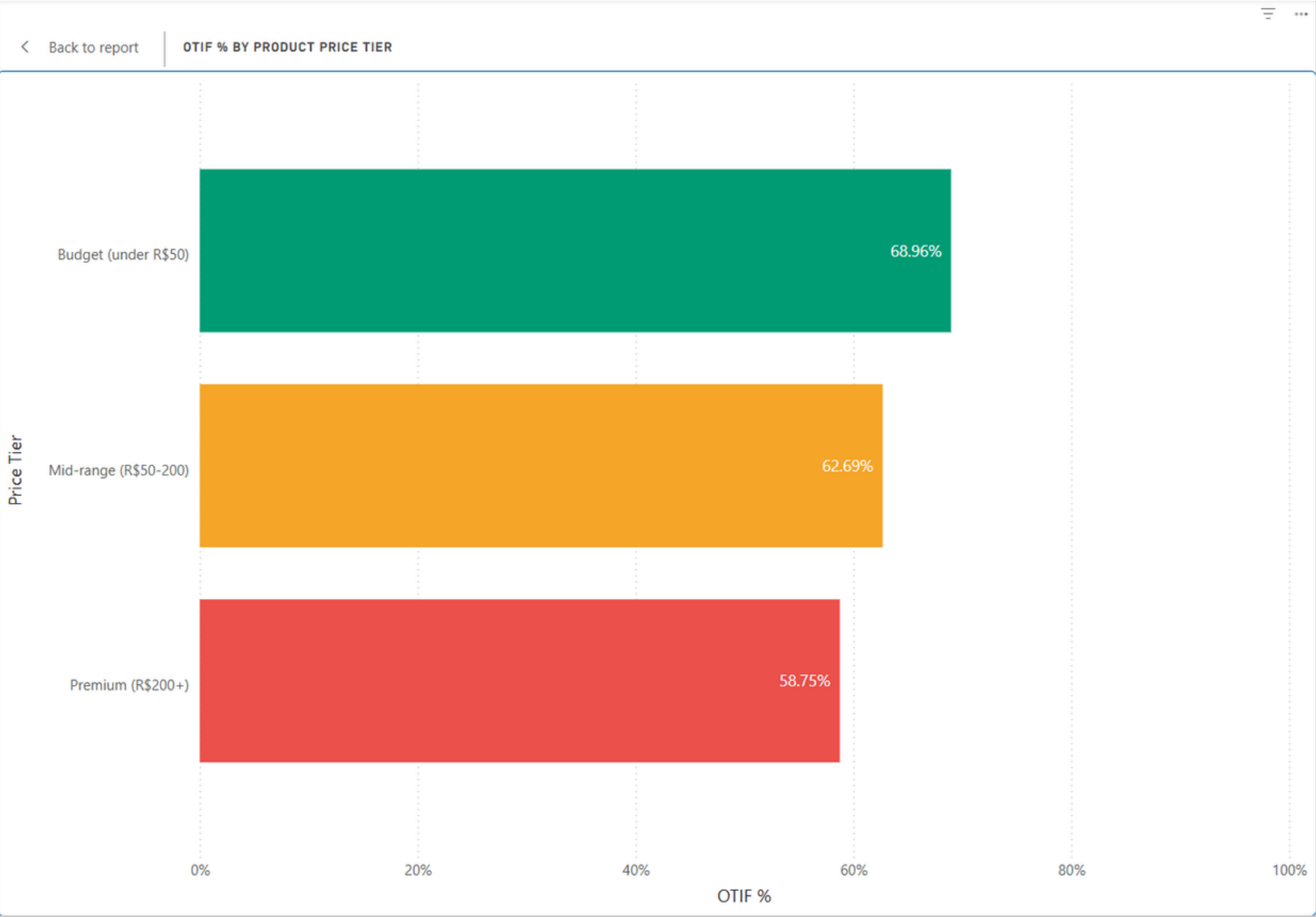
● AL State: 87.91% failure rate

● MA State: 85.63% failure rate

● AM State: 84.83% failure rate

Geographic pattern confirmed: Northern states 3x worse than São Paulo. Root cause is distance from the seller hub, driving up both failure rates and average freight costs.

Premium Products Have Worst Delivery Performance



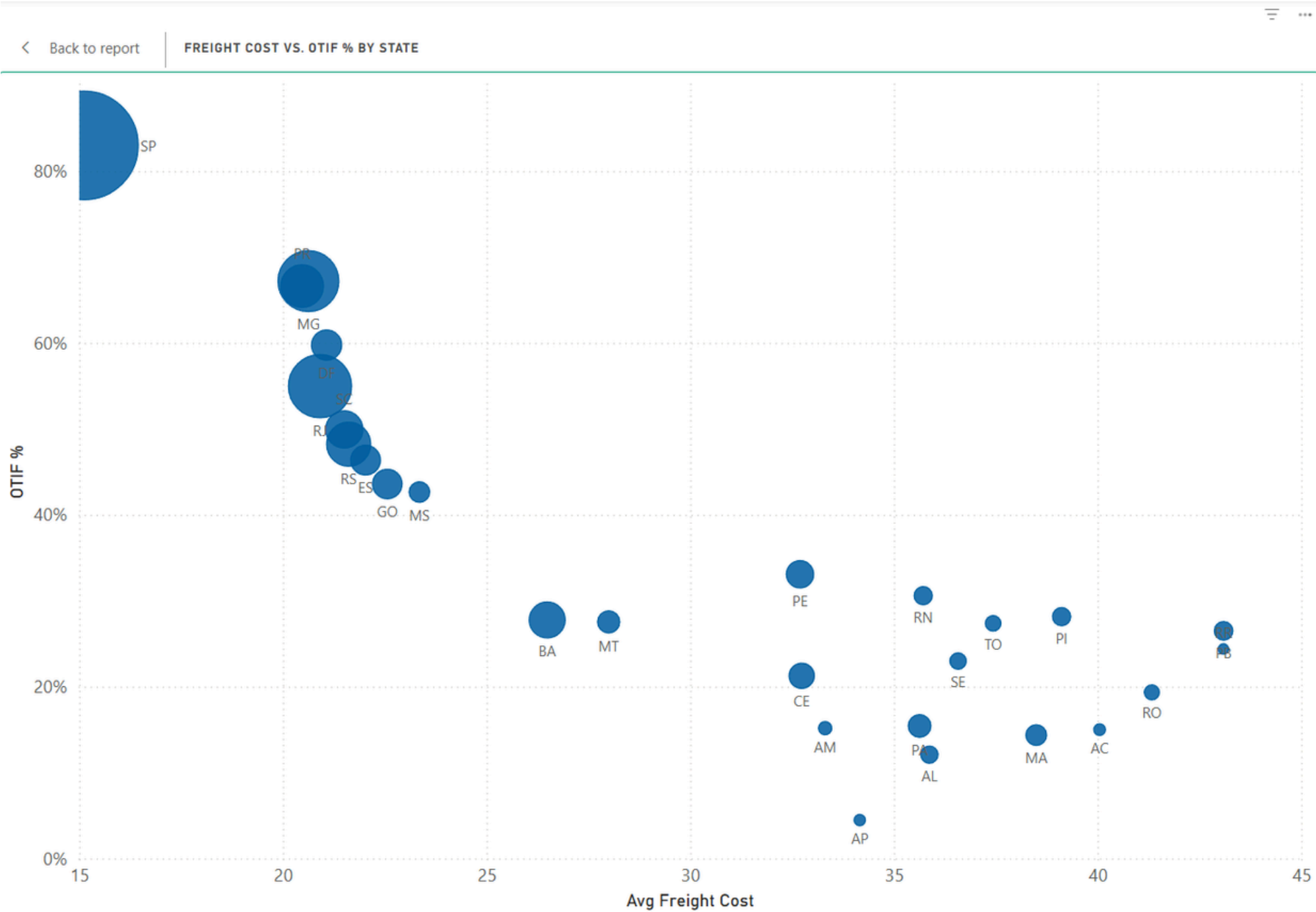
COUNTER-INTUITIVE FINDING

Premium (R\$200+): 58.75% OTIF

Budget: 68.96% OTIF

Premium customers paying MORE are receiving WORSE service. This creates double exposure on your highest LTV customers.

Predictive Risk & Freight Efficiency



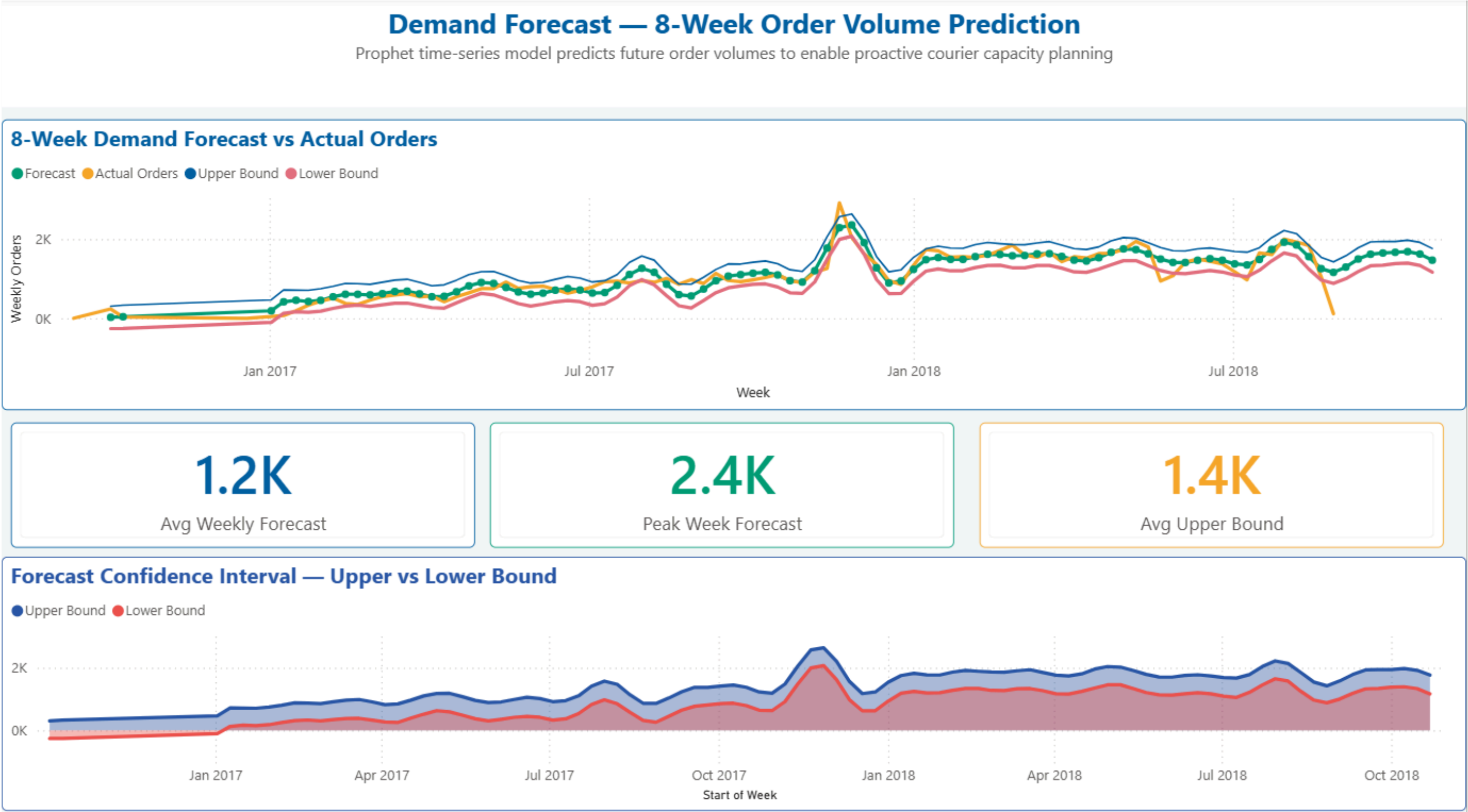
MODEL IDENTIFIES RISK :

High-risk shipments flagged via XGBoost 24 hours before failure based on carrier and weight.

SUSTAINABILITY INEFFICIENCY :

Total Volume is dominated by Maritime Shipping, but Air Freight drives the highest emissions per kg. High freight cost states (e.g., PB, RR) correlate directly with the lowest OTIF reliability.

8-Week Demand Forecast: Plan Capacity Proactively



Prophet model forecasts weekly order volumes 8 weeks ahead. Operations can pre-assign courier capacity before surges, rather than reacting after failures occur.

Three Strategic Recommendations

1

IMMEDIATE (0-30 days)

Deploy ML Risk Feed

Deploy ML risk feed to operations team. Flag high-risk shipments daily to reduce preventable failures by 15-20%.

2

SHORT TERM (1-3 months)

Reallocate Courier Capacity

Reallocate courier capacity from high-OTIF to low-OTIF northern states (AL, MA, AM).

3

MEDIUM TERM (3-6 months)

Optimize Freight & Sustainability

Shift non-urgent premium freight from highly pollutive Air/Road modes to maritime where possible, improving the 6.01 Freight Efficiency Ratio.